MICROBIOLOGY

Poststreptococcal GN & IgA Nephropathy

SGL 2 · Prof. Dr. Ruqaya M. Al Barzinji
GUE BLOCK

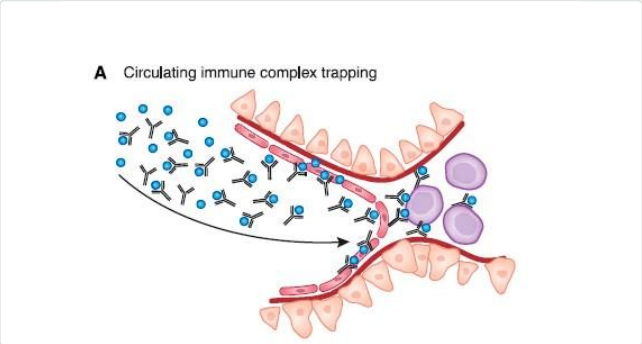
Microbiology GUE Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

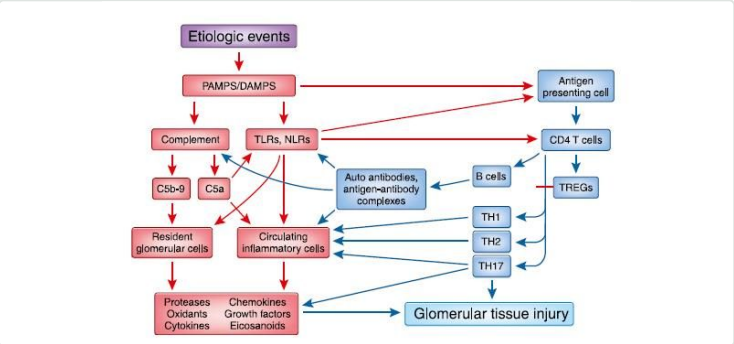
1. What antigen does **SpeB** represent in PSGN, and by which **two mechanisms** can it activate complement?
2. Which **IgA subclass**, and what specific molecular defect on it (name the deficient sugar + site), defines IgA nephropathy?
3. PSGN classically follows **strep throat** after how many days, vs a **skin infection** after how many weeks — and what distinguishes the hematuria colour/timing pattern of PSGN from IgA nephropathy?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

Glomerulonephritis (GN) = a set of renal diseases in which an **immunologic mechanism** triggers inflammation and proliferation of glomerular tissue, damaging the **basement membrane, mesangium, or capillary endothelium**. Two immune mechanisms underlie GN: (1) **deposition of circulating antigen–antibody complexes** within glomeruli (immune complex nephritis), or (2) **reaction of circulating antibodies** with antigen already trapped in the glomerulus, or with antigen that is itself part of the glomerular tissue (**anti-GBM nephritis**).



Circulating immune complex trapping in the glomerular capillary wall



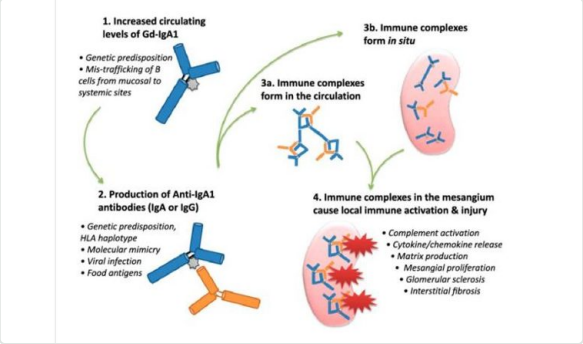
Innate + adaptive immune mechanisms converging on glomerular tissue injury

1 Illness script — Poststreptococcal glomerulonephritis (PSGN)

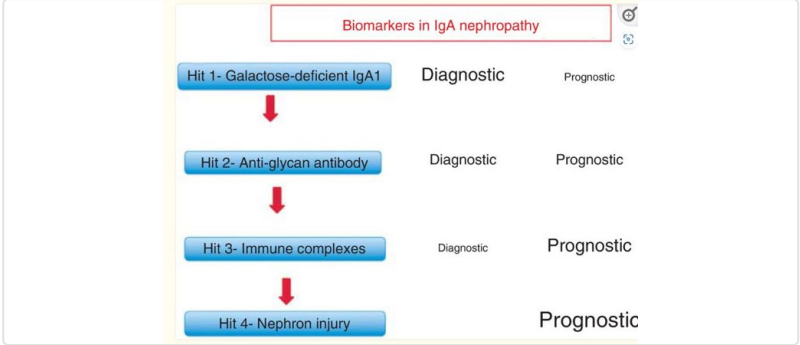
Who	Follows infection with a **nephritogenic strain of group A *Streptococcus pyogenes*** (Gram-positive bacteria)
Why	S. pyogenes secretes the exotoxin **streptococcal pyogenic exotoxin B (SpeB)** — a **cysteine protease** that cleaves various host factors and acts as a **nephritogenic antigen**; SpeB represents **90% of the secreted extracellular protein** of nephritogenic GAS strains. Antibody to SpeB **correlates with disease activity**. SpeB exhibits **molecular mimicry with endothelial cells** (hence some subendothelial deposits), can **activate complement directly** via the **mannose-binding lectin (MBL) pathway independent of IgG**, and has **plasmin-binding properties** that facilitate complement activation and may cause **proteolysis of the GBM**
What you see	**Dark, reddish-brown (tea/cola-coloured) urine** · **swelling** of face, hands and feet · **decreased urine output** · fatigue
Timing	Usually **10 days** after **strep throat/scarlet fever**; about **3 weeks** after a group A streptococcal **skin infection**
Course	Most people **recover within a few weeks** after treatment; **rare** cases progress to long-term kidney damage/kidney failure

2 Illness script — IgA nephropathy (Berger disease)

Who	**One of the commonest forms of GN**; an **autoimmune disease**. More common in young adult **males** than females, more common in **whites than blacks**, typically develops between the **late teens and late twenties** (lecture text literally reads "late thirteens" — likely a typo for twenties; **confirm with your lecturer**). **Genetic and environmental factors** play a role; **family history** is a risk factor
Why	Circulating IgA1 from affected patients is **deficient in galactose** on the **O-linked glycans of the hinge region** — called **galactose-deficient IgA1 (GD-IgA1)** — from **reduced expression of the enzyme galactosyltransferase**. GD-IgA1 is recognised as an **autoantigen** and attacked by **IgG or IgA autoantibodies**, forming **immune complexes**. These circulating complexes containing **undergalactosylated IgA1** deposit in the glomeruli and **activate mesangial cells** (modulating their function, inducing integrin expression and VEGF synthesis), driving cellular **proliferation** and overproduction of extracellular matrix, **cytokines, chemokines and complement activation → glomerular sclerosis** and **interstitial fibrosis**. Reduced **hepatic clearance** of these O-glycan–bound immune complexes raises their circulating levels further. The disease is characterised by **mesangial deposits of IgA, IgG, IgM and C3**
What you see	**Hematuria** (asymptomatic microscopic → macroscopic) · **proteinuria** (mild, may progress to heavy) · renal deficiency. About **40% have recurrent episodes of macroscopic hematuria**, often preceded **1–2 days** by an infection ("synpharyngitic hematuria")
Where it spreads / precedes	May follow **respiratory tract and GI tract infections**; also seen with **coeliac disease, hepatitis, cirrhosis, HIV**
Course	**30–40%** of patients reach **end-stage renal disease 20–30 years** after first presentation



IgA nephropathy pathogenesis — the 4-step "multi-hit" model



Biomarkers in IgA nephropathy — the same 4 hits, as diagnostic/prognostic markers

3 IgA background — why the hinge region matters

Fact	Detail
Isotypes	IgA occurs as **IgA1** (~90% of circulating IgA) and **IgA2** (~10%); produced in the **bone marrow**
Forms	Predominantly **monomeric**, with a smaller **macromolecular** fraction — **dimeric IgA (dIgA)** and **polymeric IgA (pIgA)**
Unique structural feature	IgA1 is one of the few serum glycoproteins with **O-linked oligosaccharide (O-glycan) side chains** in its **hinge region** — the site of the galactose deficiency in IgAN
Elevated levels	Plasma IgA1 is elevated in **~half** of IgAN patients — from increased bone-marrow production **and** low hepatic elimination
Consequence of altered glycosylation	Favours **self-aggregation** of IgA1 and formation of circulating IgA-containing immune complexes; increases binding of abnormally glycosylated IgA1 to **extracellular matrix**
Clearance receptor	**FcαRI/CD89**, expressed on myeloid cells — cross-linking triggers **phagocytosis, superoxide generation, antibody-dependent cellular cytotoxicity** and release of inflammatory mediators; plays a crucial role in **clearance of IgA complexes**

4 Discriminator grid — PSGN vs IgA nephropathy

Feature	PSGN	IgA nephropathy
Mechanism	**Circulating immune complex** deposition + direct complement activation by SpeB (MBL pathway)	**Mesangial IgA deposition** — galactose-deficient IgA1 recognised as autoantigen → immune complexes
Antigen	**SpeB** (streptococcal exotoxin, cysteine protease)	**GD-IgA1** (self-antigen — autoimmune)
Latency after trigger	**10 days** (throat) / **~3 weeks** (skin) — a gap exists	**1–2 days** after infection — "**synpharyngitic**" hematuria, near-simultaneous
Hematuria pattern	Dark, **tea/cola-coloured** urine, usually a single episode around the illness	**Recurrent** macroscopic hematuria in ~40% of cases, episodic with each infection
Course	Usually **self-limited**, recovery within weeks; rare progression to renal failure	**Chronic, progressive** — 30–40% reach ESRD at 20–30 years
Deposits	Subendothelial deposits (from SpeB's endothelial molecular mimicry)	**Mesangial** deposits of IgA, IgG, IgM, C3

5 Diagnosis

Modality	Finding
Clinical	Presentation pattern above ± recent infection history
Renal biopsy	**Immunofluorescence/immunohistochemistry** is the **cornerstone** of IgAN diagnosis
Urine	Hematuria with or without proteinuria
Serum	Polymeric IgA1 elevated in 30–50% of IgAN cases; **Gd-IgA1** and **glycan-specific autoantibody** levels are prime candidate diagnostic biomarkers

P Must-remember pearls

- **SpeB** is the single most important PSGN antigen — a cysteine protease, nephritogenic, complement-activating (MBL pathway, IgG-independent), and capable of GBM proteolysis via plasmin binding.
- IgA nephropathy's core lesion is **galactose-deficient IgA1 (GD-IgA1)** — the hinge-region O-glycan defect that turns IgA1 into its own autoantigen.
- **Timing is the clinical discriminator:** PSGN lags its trigger by 10 days–3 weeks; IgAN's recurrent hematuria is **synpharyngitic** — within 1–2 days of infection.
- IgA nephropathy = **Berger disease**, first described by Berger — one of the commonest causes of GN and an autoimmune disease.
- **Renal biopsy with immunofluorescence** is the diagnostic cornerstone for IgAN — mesangial IgA/IgG/IgM/C3 deposits confirm it.

Q Test yourself

1. A 9-year-old boy develops dark, tea-coloured urine, facial swelling and decreased urine output 10 days after a sore throat. What is the implicated streptococcal exotoxin, and what enzymatic activity does it have?

A. Streptolysin O; a pore-forming toxin
B. SpeB; a cysteine protease
C. DNase B; a nuclease
D. Hyaluronidase; a spreading factor
E. M protein; an adhesin, not an enzyme

2. A 19-year-old man has recurrent episodes of macroscopic hematuria, each beginning 1–2 days after an upper respiratory infection. Renal biopsy shows mesangial IgA deposits. What is the molecular defect underlying his disease?

A. Anti-GBM antibodies directly binding type IV collagen
B. Galactose-deficient IgA1 in the hinge region acting as an autoantigen
C. SpeB-mediated complement activation via the MBL pathway
D. Deposition of exogenous streptococcal antigen in the mesangium
E. A primary complement C3 deficiency

3. Which feature best distinguishes the hematuria timing of IgA nephropathy from that of PSGN relative to a preceding infection?

A. IgAN hematuria occurs weeks after infection; PSGN occurs within days
B. IgAN hematuria is synpharyngitic (1–2 days after infection); PSGN lags by 10 days to 3 weeks
C. Both occur at an identical interval after infection
D. Neither is temporally related to infection
E. PSGN hematuria recurs with each subsequent infection; IgAN does not

4. A patient with PSGN is found to have complement activation that proceeds independently of IgG. Which mechanism explains this finding?

A. SpeB directly activates complement via the mannose-binding lectin pathway
B. Anti-GBM antibodies fix complement via the classical pathway
C. IgA1 aggregates activate the alternative pathway
D. C3 nephritic factor stabilizes C3 convertase
E. Complement activation in PSGN is always IgG-dependent

5. Renal biopsy immunofluorescence in a patient with IgA nephropathy would most typically show deposits of which immunoglobulins/complement component, and in which glomerular location?

A. IgA, IgG, IgM and C3 in the mesangium
B. IgE alone in the subepithelial space
C. IgM alone along the capillary loops
D. IgG alone in a linear pattern along the GBM
E. C1q alone in the mesangium

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — SpeB (streptococcal pyogenic exotoxin B) is the key PSGN antigen; it is a cysteine protease that cleaves host factors and acts as the nephritogenic antigen.

2 — IgAN results from galactose-deficient IgA1 (GD-IgA1) — a hinge-region O-glycan defect — being recognised as a self-antigen and attacked by IgG/IgA autoantibodies, forming deposits.

3 — IgAN's recurrent macroscopic hematuria is synpharyngitic, occurring 1–2 days after infection, whereas PSGN follows its antecedent infection by a 10-day (throat) to 3-week (skin) latency.

4 — SpeB can activate complement directly through the mannose-binding lectin (MBL) pathway, independent of IgG — a distinguishing feature of its nephritogenic mechanism.

5 — IgA nephropathy is characterised by mesangial deposits of IgA, IgG, IgM and C3, demonstrated by immunofluorescence/ immunohistochemistry — the diagnostic cornerstone.

SPACED REVIEW

☐ Day 1 | ☐ Day 3 | ☐ Day 7 | ☐ Day 21 | ☐ Pre-exam